

Activation Functions: Sigmoid, Tanh and ReLU

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1. Overview

Key point: An activation function is the non-linear step inside every node. Without it a deep network is only a linear model. Sigmoid and tanh squash their input and slow deep networks down; ReLU does not squash positive values and is the default for hidden layers today.

Every node of a neural network computes a weighted sum z of its inputs plus a bias, then passes z through a function. This function, the activation function, decides how strongly the node “fires”. This Note answers three questions:

- **Why** a network needs a non-linear activation at all.
- **What** makes an activation function good: five properties.
- **How** the three classic functions, sigmoid, tanh and ReLU, score on those properties.

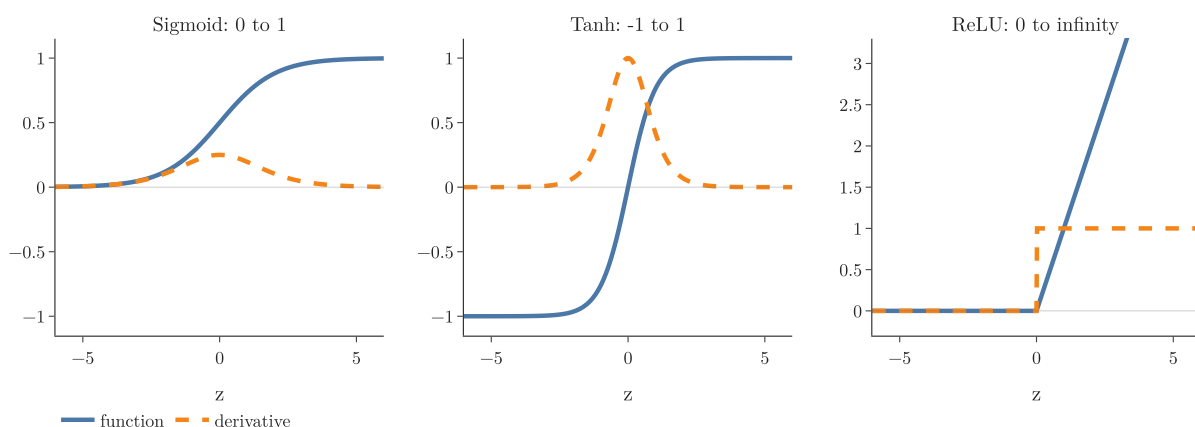


Figure 1: Sigmoid, tanh and ReLU (solid) with their derivatives (dashed). Sigmoid’s slope is never above 0.25, tanh’s never above 1; both fall to 0 for large $|z|$. ReLU’s slope is 0 or 1

Figure 1 shows the three functions side by side. Most of this Note is about what their shapes, and the shapes of their slopes, do to training.

2. Prerequisites

- The [perceptron Note](#): weighted sum, bias and activation function.
- The [sigmoid Note](#) and the [sigmoid derivative Note](#): $\sigma(z) = 1/(1 + e^{-z})$ and $\sigma'(z) = \sigma(z)(1 - \sigma(z)) \leq 0.25$.
- The [vanishing and exploding gradients Note](#): why small slopes stop early layers from learning.
- The [backpropagation how Note](#): gradients as chain-rule products.

3. What an activation function is

Key point: A node computes $z = \sum w_i x_i + b$ and outputs $a = g(z)$. The function g is the activation function: a gate deciding whether, and how much, the node is activated.

Take a network with two inputs, one hidden layer of two nodes and one output node. The first hidden node computes the weighted sum of its inputs plus its bias:

$$z = w_1 x_1 + w_2 x_2 + b_1$$

The node then outputs $a = g(z)$. The function g is the [activation function](#), also called the **transfer function**. With n inputs the output is $a = g(\sum_{i=1}^n w_i x_i + b)$.

We can picture g as a gate between what flows into a node and what flows out. The gate decides whether the node is activated and, if so, how strongly. The sigmoid and the [softmax](#) are two activation functions already met; this Note adds tanh and ReLU.

4. Why a network needs a non-linear activation

Key point: Without a non-linear activation, any number of layers collapses into one linear layer: the network can only draw a straight line, however deep it is.

4.1 An experiment: linear activations on circles

Key point: Two hidden layers of 32 nodes with linear activations reach 50% accuracy on two circles, the same as guessing. With ReLU the same network reaches 100%.

The data is `make_circles` from scikit-learn. Each **observation** (one record, one row of the data table) has two **features** (input variables) x_1 and x_2 , its position on the plane, and a **target** (the output we predict): class 1 for the inner ring, class 0 for the outer ring. There are 300 observations, 150 per class. No straight line separates the two rings. We train the same network twice for 200 epochs:

- two hidden layers of 32 nodes with `activation="linear"`, which means $g(z) = z$, the same as no activation at all;
- the same two layers with `activation="relu"`.

The output node is a sigmoid in both, because this is binary classification.

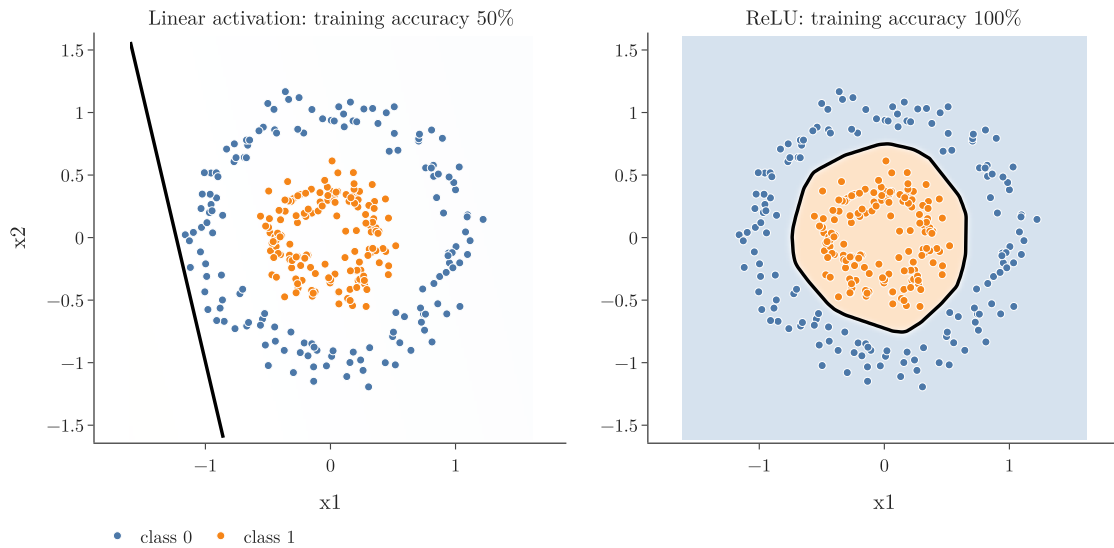


Figure 2: Decision regions after 200 epochs. Left: with linear activations the boundary (black) is a straight line, here pushed outside the data; every point gets about 0.5. Right: with ReLU the boundary bends around the inner ring

Figure 2 shows the result. The linear network ends with a loss of 0.694 and 50% training accuracy, which is exactly guessing: the best straight line it can draw is useless here. The ReLU network ends at 100%, with a boundary that bends around the inner ring.

Python: The only difference between the two models is one argument.

```
keras.layers.Dense(32, activation="linear") # no bend
keras.layers.Dense(32, activation="relu")  # bends
```

4.2 The algebra: linear layers collapse

Key point: Two linear layers $W_2(W_1a_0 + b_1) + b_2$ equal one layer $W'a_0 + b'$. The output is a degree-1 function of the input.

Stacked matrices without an activation multiply into a single matrix (see the [matrix multiplication Note](#)). The bias terms add only a shift, so they do not change that.

1. **In words:** the first layer computes $a_1 = W_1a_0 + b_1$; the second feeds that into $a_2 = W_2a_1 + b_2$; multiplying out gives one weight matrix and one bias.
2. **Formula:**

$$a_2 = W_2(W_1a_0 + b_1) + b_2 = \underbrace{W_2W_1}_{W'} a_0 + \underbrace{W_2b_1 + b_2}_{b'}$$

3. **Example:** with single numbers $W_1 = 2$, $b_1 = 1$, $W_2 = 3$, $b_2 = -1$:

$$a_2 = 3(2x + 1) - 1 = 6x + 2$$

Two layers, and the result is still a straight line in x .

The Notebook checks the collapse on the trained linear network of Figure 2. The network's three weight matrices multiply into one 2×1 matrix W' , and one layer with W' and b' gives the same predictions to within 6×10^{-8} . A network with linear activations is therefore [linear regression](#) (regression output) or [logistic regression](#) (sigmoid output), however many layers it has.

With a non-linear g , such as the sigmoid, $g(W_1 a_0 + b_1)$ is no longer a degree-1 expression. Feeding it into the next layer gives a relationship between input and output that is not a polynomial of degree 1, so the network can follow non-linear patterns. Capturing non-linear patterns is the whole reason activation functions exist.

5. Properties of an ideal activation function

Key point: A good activation function is non-linear, differentiable, cheap to compute, zero-centred and non-saturating. No classic function has all five.

5.1 Non-linear

Key point: Non-linearity is what lets a network capture non-linear patterns. Non-linearity is the one property that cannot be dropped.

The relationship between z and $g(z)$ must not be a straight line, as section 4 showed. With a non-linear activation, the [universal approximation theorem](#) says that enough nodes and layers can approximate any pattern in the data.

5.2 Differentiable

Key point: Gradient descent needs the derivative of the activation at every step, so it must exist.

Backpropagation multiplies the activation's derivative into every gradient (see the [backpropagation how Note](#)). Without a derivative there is no gradient, no update and no training.

The property is important but not strict. ReLU has no derivative at exactly $z = 0$ and is still the most used activation (section 8.2 explains how this is handled).

5.3 Computationally inexpensive

Key point: The function and its derivative are computed for every node, every observation and every epoch; a slow function makes training slow.

A network with a million nodes trained on a million observations for 100 epochs evaluates its activation $10^6 \times 10^6 \times 100 = 10^{14}$ times. Functions built from a simple comparison are faster than those built from exponentials.

5.4 Zero-centred

Key point: Outputs with mean about 0, some positive and some negative, act like normalised inputs to the next layer and make training converge faster.

A **zero-centred** activation function gives outputs whose mean over a layer is around 0. Training converges faster when the inputs are normalised (see the [standardization Note](#)), and the outputs of one layer are the inputs of the next. Tanh is zero-centred; section 6.3 shows what goes wrong when an activation is not.

5.5 Non-saturating

Key point: A saturating function squeezes any input into a fixed range, so its slope goes to 0 at the ends. The flat ends cause the vanishing gradient problem.

A **saturating function** squeezes an input from $-\infty$ to ∞ into a bounded range and flattens out there. The sigmoid squeezes into 0 to 1, tanh into -1 to 1. A **non-saturating function** has no such ceiling: ReLU, $\max(0, z)$, grows as large as its input.

Where a function is flat its slope is near 0, and a product of many near-zero slopes is the [vanishing gradient](#). The vanishing gradient appears with saturating activations, which is why non-saturation matters.

6. Sigmoid

Key point: The sigmoid gives a probability, is non-linear and differentiable; but it saturates, is not zero-centred and needs an exponential. Today it is used only in the output layer for binary classification.

6.1 Shape and slope

Key point: Output 0.5 at $z = 0$, near 0 or 1 for large $|z|$. Its derivative peaks at 0.25 and is almost 0 outside -3 to 3.

The [sigmoid](#) is $\sigma(z) = 1/(1 + e^{-z})$ (Figure 1, left). For a large z it is close to 1, for a very negative z close to 0, and $\sigma(0) = 0.5$.

Its [derivative](#) $\sigma(z)(1 - \sigma(z))$ is largest at $z = 0$, where it is 0.25. Between -3 and 3 it is reasonably large; outside, it is close to 0. At $z = 3$ it is 0.045 and at $z = 5$ only 0.0066.

6.2 Advantages

Key point: Output reads as a probability; non-linear; differentiable everywhere.

1. **Probability output.** The output lies between 0 and 1, so it can be read as a probability. For this reason we put a sigmoid in the output node for binary classification: near 1 means “yes”, near 0 means “no”.

2. **Non-linear.** The exponential makes it non-linear, so it can capture non-linear patterns.
3. **Differentiable everywhere**, with a derivative that is easy to write in terms of the function itself.

6.3 Disadvantages

Key point: Saturation causes the vanishing gradient; all-positive outputs force every weight into a node to move in the same direction; the exponential is slow.

1. **Saturating, so the gradient vanishes.** The update is $w_{\text{new}} = w_{\text{old}} - \eta \partial L / \partial w$. If z is outside about -3 to 3 , the sigmoid's slope is near 0, the gradient is near 0 and $w_{\text{new}} \approx w_{\text{old}}$: no training. In a deep network these small slopes multiply from layer to layer (see the [vanishing gradients Note](#)).

z becomes large in two ways:

- **Large inputs.** Usually not the cause, because we scale the inputs first.
- **Large weights and biases.** If the starting weights are big, z is big and the node starts saturated (see the [weight initialisation Note](#)).

Saturation is the main reason the sigmoid has disappeared from hidden layers (Goodfellow et al. 2016, §6.3.2).

2. **Not zero-centred, so training is slow.** Every sigmoid output is positive. Take two sigmoid nodes, with outputs a_{21} and a_{22} , feeding one output node through weights w_{21} and w_{22} :

$$z_{31} = w_{21} a_{21} + w_{22} a_{22} + b_{31}$$

1. **In words:** by the chain rule, the gradient of each weight is a factor shared by both, times that weight's input.

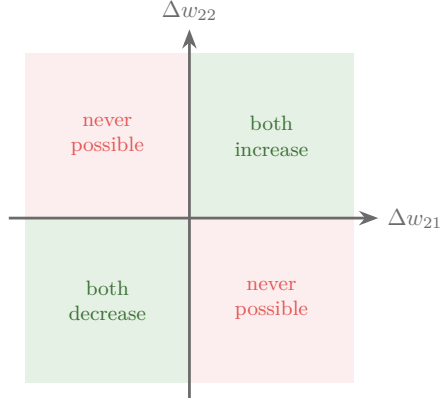
2. **Formula:** with the shared factor $\delta = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_{31}}$,

$$\frac{\partial L}{\partial w_{21}} = \delta \cdot a_{21}, \quad \frac{\partial L}{\partial w_{22}} = \delta \cdot a_{22}$$

3. **Example:** with $\delta = -0.5$ and sigmoid outputs $a_{21} = 0.6$, $a_{22} = 0.3$, the gradients are -0.30 and -0.15 : both negative. Had a_{22} been a tanh output of -0.3 , they would be -0.30 and $+0.15$.

Because a_{21} and a_{22} are both positive, both gradients take the sign of δ . So in one update the weights into a node either all increase or all decrease; they can never move in opposite directions.

(a) Directions one update can take



(b) Reaching the best weights

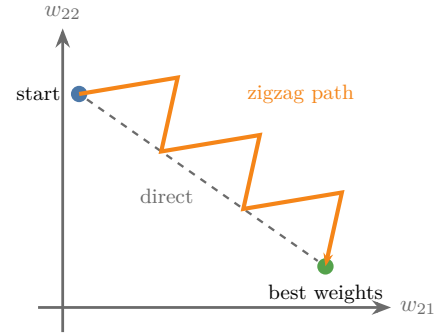


Figure 3: (a) With all-positive inputs, one update moves every weight into a node the same way: only the green quadrants are possible. (b) When the best weights need one weight up and the other down, gradient descent has to zigzag

Figure 3 shows the consequence. If reaching the minimum needs w_{21} to grow while w_{22} shrinks, no single step can do that. Gradient descent zigzags, like driving to a place on our left when we are not allowed to turn left, and it converges slowly (LeCun et al. 1998).

The Notebook checks the sign rule with one hidden layer of 10 nodes feeding a sigmoid output, for one observation. With sigmoid hidden nodes, the 10 gradients of the output node’s weights all have the same sign (+++++). With tanh hidden nodes they are mixed (-+-+-+).

3. Computationally expensive. Each evaluation needs an exponential. In the Notebook, 10 million sigmoids take about 650 ms, against 135 ms for ReLU.

In short, the sigmoid is rarely used in hidden layers today. The sigmoid stays in the output layer for binary classification.

7. Tanh

Key point: Tanh is a stretched sigmoid with range -1 to 1 . Tanh is zero-centred, which fixes the sigmoid’s slow training, but it still saturates and still needs exponentials.

7.1 Shape and slope

Key point: $\tanh(z) = (e^z - e^{-z}) / (e^z + e^{-z})$, with derivative $1 - \tanh^2(z)$, peaking at 1 .

1. **In words:** the **hyperbolic tangent (tanh)** is an S-shaped curve like the sigmoid, but from -1 to 1 , passing through 0 at $z = 0$.

2. **Formula:**

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}, \quad \tanh'(z) = 1 - \tanh^2(z)$$

3. **Example:** at $z = 1$, $e^1 = 2.718$ and $e^{-1} = 0.368$, so

$$\tanh(1) = \frac{2.718 - 0.368}{2.718 + 0.368} = \frac{2.350}{3.086} = 0.762, \quad \tanh'(1) = 1 - 0.762^2 = 0.42$$

Like the sigmoid, its derivative is written in terms of the function itself. The derivative's peak is 1 instead of 0.25, and the curve is steeper (Figure 1, middle).

Extra: Tanh is a rescaled sigmoid: $\tanh(z) = 2\sigma(2z) - 1$. To see it, multiply the top and bottom of $2\sigma(2z) - 1$ by e^z :

$$2\sigma(2z) - 1 = \frac{2}{1 + e^{-2z}} - 1 = \frac{1 - e^{-2z}}{1 + e^{-2z}} = \frac{e^z - e^{-z}}{e^z + e^{-z}} = \tanh(z)$$

The Notebook confirms the identity numerically. The steeper middle of tanh also means tanh flattens sooner: at $z = 3$ its slope is 0.0099, against the sigmoid's 0.045.

7.2 Advantages and disadvantages

Key point: Gains zero-centred outputs; keeps saturation and exponentials.

Advantages:

1. **Non-linear**, so it captures non-linear patterns.
2. **Differentiable** everywhere.
3. **Zero-centred**. Outputs are both positive and negative. The gradients of the weights into a node can then have different signs, and training is faster than with the sigmoid (LeCun et al. 1998, §4.4). In the Notebook, over all 300 observations, a 128-node tanh layer has mean output 0.000 with half the values positive, while a sigmoid layer has mean 0.500 with every value positive.

Disadvantages:

1. **Saturating**. For large $|z|$ the slope is 0, so tanh also suffers from the vanishing gradient.
2. **Computationally expensive**, because of the exponentials (about 270 ms for 10 million values in the Notebook).

Tanh fixed the sigmoid's slow training from non-zero-centred outputs, but not the vanishing gradient.

8. ReLU

Key point: ReLU, $\max(0, z)$, is non-linear, does not saturate for positive z , is cheap and converges faster than sigmoid and tanh. Its weaknesses: no derivative at 0, not zero-centred, and dying nodes.

ReLU (rectified linear unit) is $\max(0, z)$: 0 for negative z , z itself for positive z (Figure 1, right). ReLU is the most used activation in hidden layers today, the recommended default (Goodfellow et al. 2016, §6.3).

8.1 Advantages

Key point: Non-linear (two ReLUs already make a bend), non-saturating on the positive side, no exponential, faster convergence.

1. Non-linear. ReLU looks like two straight pieces, which makes some people think it is linear. A linear function would be $f(z) = z$ everywhere; $\max(0, z)$ has a corner, so it is not. Combining ReLUs builds more complex shapes:

- 1. In words:** subtract one shifted ReLU from another.
- 2. Formula:** $f(x) = \max(0, x + 1) - \max(0, x - 1)$
- 3. Example:** $f(-2) = 0 - 0 = 0$; $f(0) = 1 - 0 = 1$; $f(3) = 4 - 2 = 2$. Flat, then rising, then flat again.

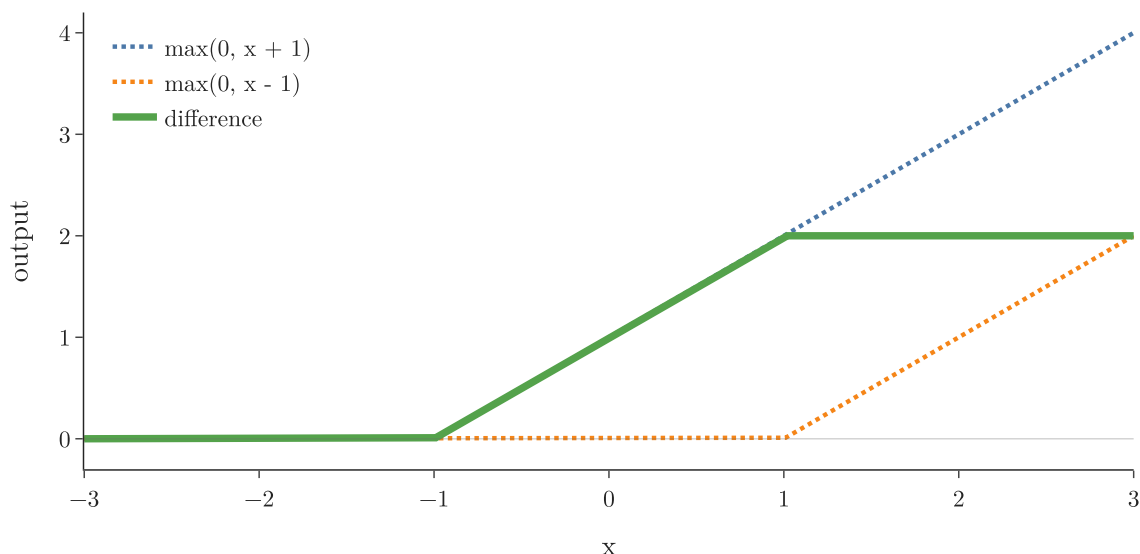


Figure 4: Two ReLUs (dotted) and their difference (green): flat, rising, flat. No straight line has two corners

Figure 4 shows the result: a curve with two corners. The ReLU boundary in Figure 2 is built the same way, from straight pieces joined at corners.

2. Non-saturating in the positive region. For positive z the output grows without limit and the slope stays 1, so the gradient does not vanish there.

3. Computationally inexpensive. Computing it is one comparison, and its derivative is 0 or 1. No exponentials.

4. Faster convergence than sigmoid and tanh. On the CIFAR-10 images, a four-layer ReLU network reached a 25% training error about six times faster than the same network with tanh (Krizhevsky et al. 2012, §3.1, Figure 1).

8.2 Disadvantages

Key point: No derivative at exactly 0 (we just pick one), outputs never negative (batch normalisation helps), and nodes that can die.

1. Not differentiable at $z = 0$. The two pieces meet at a corner, which has no single slope. In code we simply choose one: slope 0 for $z < 0$, slope 1 for $z > 0$, and one of the two at exactly 0.

Extra: TensorFlow uses slope 0 at exactly $z = 0$ (the Notebook's gradient check returns 0 there); the convention above, 1, is the other common choice. The choice hardly matters in practice: of the 19,200 weighted sums in the two hidden layers of the trained ReLU network of Figure 2, none is exactly 0.

2. Not zero-centred. Like the sigmoid's, ReLU's outputs are never negative, with the same effect on the gradient signs as in section 6.3. [Batch normalisation](#) addresses this: it normalises the values passed between layers.

3. The dying ReLU problem. A node whose z stays negative outputs 0 with slope 0, so it stops learning (see section 6.2 of the [vanishing gradients Note](#)). The causes and the fixes are in the [ReLU variants Note](#).

Python: Choosing activations in Keras.

```
model = keras.Sequential([
    keras.Input(shape=(2,)),
    keras.layers.Dense(32, activation="relu"), # hidden: ReLU
    keras.layers.Dense(32, activation="tanh"), # or tanh
    keras.layers.Dense(1, activation="sigmoid")] # output
```

The strings "sigmoid", "tanh", "relu" and "linear" pick the function for every node of that layer.

9. Summary

Property	Sigmoid	Tanh	ReLU
Formula	$1/(1 + e^{-z})$	$(e^z - e^{-z})/(e^z + e^{-z})$	$\max(0, z)$
Range	0 to 1	-1 to 1	0 to ∞
Largest slope	0.25	1	1
Non-linear	yes	yes	yes
Differentiable	everywhere	everywhere	except at 0
Zero-centred	no	yes	no
Saturating	yes	yes	not for $z > 0$
Cost (10 million values)	about 650 ms	about 270 ms	about 135 ms
Used today	output, binary classification	some hidden layers	most hidden layers

- Without a non-linear activation, any network is a linear model (50% vs 100% on the circles).
- An ideal activation is non-linear, differentiable, cheap, zero-centred and non-saturating.
- Saturating functions (sigmoid, tanh) cause the vanishing gradient.
- All-positive outputs (sigmoid, ReLU) make the weights into a node move together, so training zigzags.
- ReLU is the default for hidden layers; its dying-node problem leads to the ReLU variants.

10. Sources

- LeCun, Y., Bottou, L., Orr, G. B. and Müller, K.-R. (1998). Efficient BackProp. In *Neural Networks: Tricks of the Trade*, Springer, section 4.3 (all-positive inputs make the weight updates zigzag) and section 4.4 (symmetric sigmoids such as tanh converge faster than the logistic sigmoid).
- Krizhevsky, A., Sutskever, I. and Hinton, G. E. (2012). ImageNet Classification with Deep Convolutional Neural Networks. NeurIPS, section 3.1 (ReLU trains several times faster than tanh).
- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*, MIT Press, section 6.3 (ReLU as the default hidden unit; sigmoid hidden units discouraged because they saturate).

11. Key terms

Term	Meaning
Transfer function	Another name for the activation function
Linear activation	$g(z) = z$, the same as no activation; Keras' <code>activation="linear"</code>

Term	Meaning
Zero-centred activation	An activation whose outputs average about 0, positive and negative
Saturating function	A function that squeezes any input into a bounded range, so its slope goes to 0 at the ends
Non-saturating function	A function with no upper limit on its output, such as ReLU for positive inputs
Tanh (hyperbolic tangent)	The activation $(e^z - e^{-z})/(e^z + e^{-z})$, an S-curve from -1 to 1 with derivative $1 - \tanh^2(z)$
Observation	One record of the data: one row of the data table
Feature	An input variable: one column of the data table, such as x_1
Target <code>make_circles</code>	The output we predict, such as the class scikit-learn function that generates two concentric rings of observations, one ring per class